Wing and Tail De-icing System (Mod S.O.O. 6004 and Mod 6/2042) – General Data

1. General

The electrically controlled wing and tail de-icing system utilizes bleed air tapped from the engine compressors to inflate and deflate rubber de-icing boots installed on the leading edges of both wings and the horizontal stabilizer. The inflation and deflation pulses flex both wing boot inner sections, both wing boot outer sections, and the left and right boots on the horizontal stabilizer, in sequence, to break up any ice formation. Two wing inspection lights may be installed which can be used during darkness in icing conditions, to determine the thickness of ice accumulation on the wing leading edges (refer to 33-40-00, Exterior Lights (Mod 6/2058) – General Data).

For aircraft incorporating S.O.O. 6367 (Wing De-ice Provisions Removal), the wing de-icing tube assembly is not installed.

2. Description

The system consists of three distributor valves, an ejector, an electronic timer, three water separators, control switches, and rubber boots bonded to the leading edge of each wing and horizontal stabilizer. The regulated air supply is provided from a pneumatic system package (refer to 36-00-00, Pneumatic – General Data), located in the cabin roof. Electrical power for the wing and tail de-icing system is derived from the left and right DC bus bars through two circuit breakers marked AIRFRAME DEICE AUTO, and AIRFRAME DEICE MANUAL, located on the circuit breaker panel. If sufficient pressure exists at the pressure sensing switch for each horizontal stabilizer boot, the message “R STAB” or “L STAB” shall appear in green inverse video in the de-ice portion of the electrical system pane. An electrical heater jacket is installed around each distributor valve (Mod 6/1440). Refer to Figure 1 for details.

Included in the pneumatic system package is a low-pressure switch, the pressure switch consists of a single switch that activates when pneumatic pressure has decreased to 13 – 15 psi. A CAS message is displayed to the crew when pneumatic pressure falls below this level and the surface de-ice system has been selected ON. This message indicates that there is insufficient pressure to satisfactorily operate the de-ice boots.

A. Distributor Valves

The distributor valves are mounted one in each wing at station 260.00, and one in the rear section of the fuselage at station 486.00 approximately. Each distributor valve incorporates a pressure inlet port, a suction outlet port, two ports A and B to the de-icer boots, an exhaust port piped overboard to a low-pressure area, and two control solenoids, A and B.

B. Ejector

The ejector, located in the rear section of the fuselage, aft of the horizontal stabilizer distributor valve, is a venturi-type device which creates a vacuum to deflate the de-icing boots. The ejector is in continuous operation whenever the pneumatic supply is switched on.

C. Electronic Timer

The electronic timer, mounted in the cabin roof, is an electronic device which provides a cycling electrical output to the de-icing distributor valve solenoids. Fast or slow cycling of the timer may be selected by the operation of a switch, mounted on the de-icing control panel on the right hand switch panel.

D. Water Separators

Three water separators, one installed in each wing pneumatic line, and a third installed in the pneumatic line to the horizontal stabilizer distributor valve, are of the self-draining type. Water separators are located adjacent to the distributor valves (Mod 6/1440) and each separator contains a disposable 2 micron filter.

E. De-icing Boots

Inflatable de-icing boots are bonded to the leading edges of each wing, from a point just outboard of the engine nacelles, to the wing tips. Similar type boots are bonded to the entire leading edge of the horizontal stabilizer. Each boot is connected by hoses to the A and B ports of its relevant distributor valve: inner wing to A, outer wing to B, left stabilizer to A and right stabilizer to B.

F. Control Switches

The control switches are located on the de-icing panel located in the right pilot's sub-panel, under the label SURFACE DE-ICE. They consist of the following:

1) A BOOTS AUTO switch for the selection of automatic speeds or manual, with the positions marked FAST, SLOW and OFF/MANUAL.

2) A manual mode switch to momentarily inflate the wing de-icer boots. Alternate positions are marked INNER and OUTER.

3) A second manual STABILIZER switch to momentarily inflate the horizontal stabilizer boots. Alternate positions are marked LEFT and RIGHT.

4) A switch marked VALVE HEAT (Mod 6/1440), to energize the three distributor valve heaters.

G. Horizontal Stabilizer De-icing Boot Indication

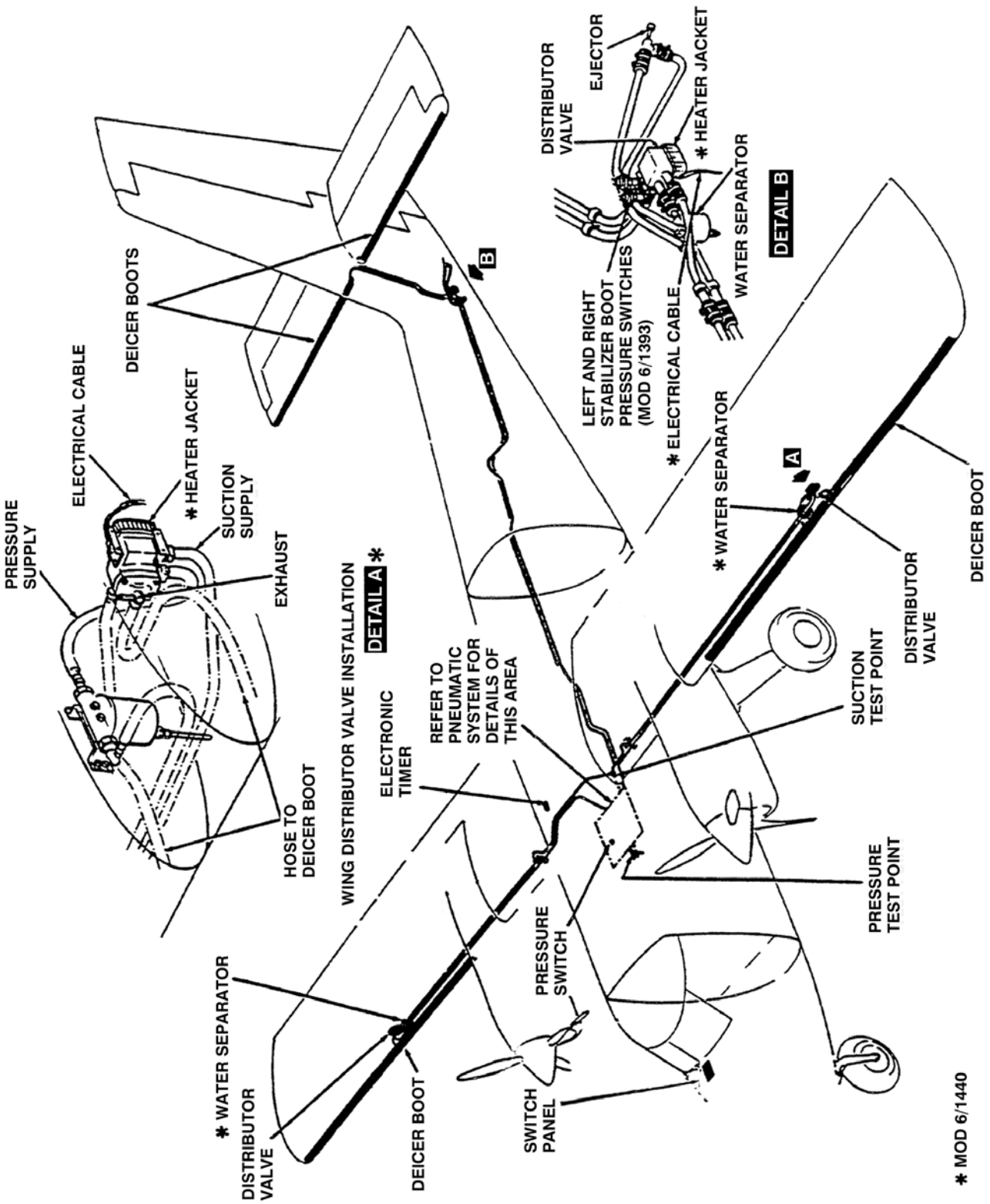
The Horizontal Stabilizer De-icing Boot Indicator is generated by a pressure switch in the air pressure line to the related tail plane de-icer boot. The message is displayed in the Fuel System pane: “L STAB” and “R STAB” is illuminated for the period of time that air at operating pressure is metered to each boot in each inflation cycle, in either automatic or manual mode.

H. Horizontal Stabilizer De-icing Pressure Switches (Mod 6/1393)

The two pressure switches are installed in the rear fuselage at the distributor valve pressure outlets A and B. The normally open contacts of the pressure switches close at 15 psi rising pressure.

I. Distributor Valve Heater Jacket (Mod 6/1440)

An electrical heater jacket is wrapped around the solenoid body of each distributor valve. The heater jacket is secured to the distributor valve by two springs. All three heater jackets are powered through the AIRFRAME DEICE MANUAL circuit breaker and VALVE HEAT switch.



Wing and Tail De-icing Systems (Mod S.O.O. 6004)
Figure 1

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3. Operation

With the system circuit breakers closed and the BOOTS AUTO switch set to FAST or SLOW, the regulated pneumatic supply (refer to 36-10-00, Distribution (Mods S.O.O. 6004) – General Data) is directed to the distributor valves and the ejector. The venturi action of the ejector provides a suction at each of the distributor valves. The distributor valve solenoids are energized in a sequence determined by the operation of the electronic timer, at a fast or slow rate depending on the selection of the BOOTS AUTO, FAST or SLOW switch. As each solenoid is energized, a servo valve is operated to reverse suction and pressure conditions across a diaphragm within the distributor valve. The diaphragm mechanically positions a valve which shuts off the suction and admits pressure into the boot port, thereby allowing a section of the boot to inflate for a predetermined time. When the solenoid is deenergized, air flow through the valve on that side is cut off, and air discharges out of that particular boot section through a check valve until the pressure drops to approximately 1 inch Hg, at which time, that boot section is ported to the suction system and the remaining air is evacuated. The boot is held down until another cycle starts. When the system is selected OFF, only suction is applied to the de-icer boots to deflate and hold them flush to the wing profile. In the event of a failure of the electronic timer, the selector switch can be selected to OFF/MANUAL, and the operation of the distributor valves controlled by momentarily operating the WING INNER–OUTER or STABILIZER LEFT–RIGHT switches. A L STAB or R STAB indication comes on whenever the associated pressure switch contacts close at 15 psi rising pressure. On aircraft incorporating Mod 6/1440, when the VALVE HEAT switch is selected ON, the heater jacket of each distributor valve is energized. Refer to Figure 1 and PSM 1-64-2W for electrical schematic.

4. Adjustment/Test

Refer to Figure 1 for details.

A. Function Test Wing and Tail De-icing System

Equipment required:

- Pressure rig and gauge required for testing the de-icing system, is as specified for the pneumatic and air conditioning systems (refer to 21-40-00, Heating System – General Data and 36-10-11, Pressure Regulator Valve – Maintenance Practices).
- Suction gauge capable of measuring 4 to 10 inches Hg, with the necessary adapter to connect gauge to the suction test point.

1) Remove cabin roof upholstery in location of station 177.00 to provide access to heat exchanger.

2) Disconnect bleed air line at inlet to heat exchanger and connect pressure rig.

3) Remove cabin roof upholstery as required and connect suction gauge to the tee fitting in suction line forward of station 200.36.

4) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

5) With de-icer circuit breakers (AIRFRAME DEICE AUTO and AIRFRAME DEICE MANUAL) engaged, check that the low pressure advisory message is displayed.

6) Start pressure rig and pressurize system to maintain 18 ± 2 psi on rig pressure gauge, and 4 to 10 inches Hg suction. Check that low pressure advisory message is no longer displayed.

Note If a leak in the system is apparent, leak test the system in accordance with 36-10-11, Pressure Regulator Valve – Maintenance Practices.

7) Select system selector BOOTS AUTO switch to FAST. A temporary loss of pressure and suction is allowed.

8) Check that sequence of boot inflation, correct inflation time and dwell time are in accordance with Table 1 — De-icer Boots Timing Sequence. Check the left and right horizontal stabilizer boot indicators each come on for 3 seconds in each 60 second cycle (Mod 6/1393). Ensure that the indicators do not, at any time, come on together.

9) Ensure that each boot section deflates fully in 20 seconds.

10) Select cycle selector switch to BOOTS AUTO and SLOW and check that sequence of inflation, correct inflation time and dwell time are in accordance with Table 1 — De-icer Boots Timing Sequence.

11) Select BOOTS AUTO switch to OFF/MANUAL.

12) Select each distributor valve switch, in turn, to WING INNER, WING OUTER, LEFT STABILIZER, RIGHT STABILIZER. The appropriate boot section must inflate in time specified in Table 1 — De-icer Boots Timing Sequence, remain inflated as long as switch is held in position, and deflate when switch is released.

13) Check that when LEFT is selected the L STAB indicator comes on within 2 seconds, and that when RIGHT is selected the R STAB indicator comes on within 2 seconds. Ensure that the indications do not, at any time, come on together.

14) Release pressure and note that Pneumatic Press Low advisory message is displayed at 15 ± 2 psi. Ensure that both L STAB and R STAB indications are OFF.

15) Select VALVE HEAT (Mod 6/1440), switch ON and check that heater jackets on wing and fuselage distributor valves are warm. Set VALVE HEAT switch to OFF.

16) Set BOOTS AUTO switch to OFF/MANUAL.

17) Disconnect test rig and suction gauge. Reconnect bleed air line at inlet to heat exchanger. Reinstall cabin roof upholstery where necessary.

18) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Table 1 De-icer Boots Timing Sequence

SEQUENCE OF INFLATION	TIME TAKEN FOR INFLATION	DWELL TIME AT INFLATION SELECTION	
		FAST	SLOW
Wing Inner	5 seconds	44 ± 10 seconds	2 min 44 ± 20 seconds
Wing Outer	5 seconds		
Left Stabilizer	3 seconds		
Right Stabilizer	3 seconds		

B. Operational Test (Pre Mod 6/2314)

1) Set BUS TIE switch to NORMAL and POWER SOURCE switch to BATTERY.

2) Select MASTER switch to ON.

3) Momentarily select VALVE HEAT switch to ON and check that Battery Amps Indication shows a discharge, then set switch to OFF.

4) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.) and set POWER SOURCE switch to EXTERNAL.

5) Start engine. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

6) Ensure AIRFRAME DEICE AUTO and AIRFRAME DEICE MANUAL circuit breakers are engaged and Pneumatic Press Low advisory message is extinguished.

7) Select BOOTS AUTO switch to FAST.

8) Have second operator outside aircraft to observe boot inflation sequence as listed in Table 1 — De-icer Boots Timing Sequence. Check horizontal stabilizer de-icing boots indicator operation as in Para 4.A.8).

9) Select BOOTS AUTO switch to SLOW and repeat Para 4.B.8).

10) Select switch BOOTS AUTO to OFF/MANUAL.

11) Select each distributor valve switch in turn, to WING INNER, WING OUTER, LEFT STABILIZER, RIGHT STABILIZER. The appropriate boot section must inflate in time specified in Table 1 — De-icer Boots Timing Sequence, remain inflated as long as switch is held in position, and deflate when switch is selected to OFF or another position. Check horizontal stabilizer de-icing boots indicator operation as in Para 4.A.13).

12) Select BOOTS AUTO switch to OFF/MANUAL.

13) Shutdown engine (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data) and check Pneumatic Press Low caution CAS message is displayed before switching MASTER switch to OFF.

14) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. Operational Test (Post Mod 6/2314)

1) Set BUS TIE switch to NORMAL and BATTERY switch to ON.

2) Momentarily select VALVE HEAT switch to ON and check that Battery Amps Indication shows a discharge, then set switch to OFF.

3) Set BATTERY switch to OFF.

4) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.) and set EXTERNAL switch to ON.

5) Start engine. (Refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

6) Ensure AIRFRAME DEICE AUTO and AIRFRAME DEICE MANUAL circuit breakers are engaged and Pneumatic Press Low advisory message is extinguished.

7) Select BOOTS AUTO switch to FAST.

8) Have second operator outside aircraft to observe boot inflation sequence as listed in Table 1 — De-icer Boots Timing Sequence. Check horizontal stabilizer de-icing boots indicator operation as in Para 4.A.8).

9) Select BOOTS AUTO switch to SLOW and repeat Para 4.C.8).

10) Select switch BOOTS AUTO to OFF/MANUAL.

11) Select each distributor valve switch in turn, to WING INNER, WING OUTER, LEFT STABILIZER, RIGHT STABILIZER. The appropriate boot section must inflate in time specified in Table 1 — De-icer Boots Timing Sequence, remain inflated as long as switch is held in position, and deflate when switch is selected to OFF or another position. Check horizontal stabilizer de-icing boots indicator operation as in Para 4.A.13).

12) Select BOOTS AUTO switch to OFF/MANUAL.

13) Shutdown engine (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data) and check Pneumatic Press Low caution CAS message is displayed before switching EXTERNAL switch to OFF.

14) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).